Contrastive Self-Supervised Learning-Based Background Reconstruction for Hyperspectral Anomaly Detection

Xiaoming Sun, Yuxiang Zhang, *Senior Member, IEEE*, Yanni Dong, *Senior Member, IEEE*, and Bo Du, *Senior Member, IEEE*

***Abstract*— Deep learning (DL) has received a lot of attention in hyperspectral anomaly detection (HAD) in recent years. While some progress has been made in boosting the generalization of DL-based HAD methods, existing methods still face limitations in labeled sample generation or transfer flexibility. To address these issues, we propose a contrastive self-supervised learning-based background reconstruction method for HAD (CSSBR). By constructing a self-supervised pretraining model based on a pixel- and patch-level masking strategy and a dual attention network (DAN) encoder, the pretraining model can learn general background representation without labeled sample generation. The DAN encoder is constructed by the vision transformer (ViT) and channel attention module (CAM) to extract the global context information and the correlation between spectra. A transfer learning model is constructed based on the DAN encoder and a lightweight decoder, which is simply fine-tuned by background reconstruction guided (BRG) loss to flexibly transfer the pretrained model to various HAD tasks. Experiments on four challenging hyperspectral datasets confirm that the CSSBR method outperforms other state-of-the-art methods.**

***Index Terms*— Background reconstruction, contrastive learning, hyperspectral anomaly detection (HAD), self-supervised learning (SSL).**

I. Introduction

HYPERSPECTRAL imaging technology records both spectral and spatial information of ground objects [1]. Compared with other remote sensing images, hyperspectral images (HSIs) contain hundreds or even thousands of almost continuous spectral bands, which can provide very detailed spectral information [2]. Therefore, HSI can provide unique advantages for anomaly detection tasks. Hyperspectral anomaly detection (HAD) focuses on distinguishing unusual materials from typical backgrounds in the absence of a priori knowledge of anomalous targets or backgrounds [3]. In recent years, HAD has flourished and is widely used in medicine [4], oil spill detection [5], civilian rescue [6], and other fields [7].

Over the past few decades, numerous traditional methods for HAD have been proposed. The most classical of these methods are the statistical assumption-based methods, which assume that the backgrounds obey a certain statistical distribution and consider the pixels that deviate significantly from this distribution as anomalies. The Reed–Xiaoli (RX) detector [8] is the most widely used algorithm among the statistical assumption-based methods, which assumes that the backgrounds obey a multivariate Gaussian distribution and utilizes all the pixels in the image to compute the background statistics. According to the different ways of background statistical estimation, various RX-based variants have been proposed, such as kernel RX [9], subspace RX [10], local RX [11], local adaptive iterative RX (LAIRX) [12], and two-step generalized likelihood ratio test (2S-GLRT) [13]. Subsequently, the representation-based methods have gradually garnered widespread attention, encompassing collaborative representation [14], [15], [16], low-rank representation [17], [18], [19], tensor representation [20], [21], and entropy representation [22]. For example, Li and Du [14] proposed the collaborative representation-based detector (CRD) to represent the pixel with the local background dictionary and separate anomalies from backgrounds by the recovery residuals. Ma et al. [23] combined the integrating fractional Fourier transform (FrFT) with the row-constrained low-rank and sparse matrix decomposition technique. Wang et al. [20] extended the 3-D tensor low-rank model to separate the background component. Tao et al. [22] combined FrFT and Shannon entropy to maximize the distinction between anomalies and backgrounds. These traditional methods are generally simple and extract shallow features to distinguish anomalies from backgrounds.

Recently, deep learning (DL) has gained prominence in HAD due to its powerful capabilities of feature mining and feature representation. The most widely used deep neural networks include convolutional neural network (CNN) [24], [25], [26], autoencoder (AE) [27], [28], [29], and generative adversarial network (GAN) [30], [31], [32]. For example, Li et al. [33] proposed a two-stream CNN with deep support vector data description that incorporates feature learning and

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anomaly detection into a unified framework. Wang et al. [34] proposed an autonomous HAD network based on fully convolutional AE (Auto-AD), which designs the adaptive weighted loss function and the anomaly detector in an end-to-end manner. Xiang et al. [35] enhanced the ability of representing the background by introducing a guided model to the hidden layer of AE. Xie et al. [36] proposed a spectral adversarial feature learning method for HAD, which uses the adversarial AE (AAE) as its backbone and introduces the structure tensor matrix to adaptively calculate the contribution rate of each feature. Wang et al. [37] proposed an HAD method based on variational background inference and GAN (VBIGAN-AD), which establishes the relationship between data samples and latent samples through two subnetworks to capture the data distribution. It is worth noting that self-supervised learning (SSL) with deep neural networks is gaining attention in the field of HAD. Wang et al. [38] proposed a pixel-shuffle downsampling masked-spot reconstruction network (PDBSNet) in a self-supervised operating fashion for HAD, which achieves effective background reconstruction by means of an AE consisting of centrally masked and dilated convolutions. Liu et al. [39] proposed a self-supervised multiscale network (MSNet) for HAD with enhanced separation training, where the network consists of convolutional AEs at multiple scales. Wang et al. [40] proposed a self-supervised HAD transformer (SSHADFormer) method based on finite spatialwise attention and AE, which performs background reconstruction via the RGB cube associated with the given dataset. The DL-based methods described above are usually trained and tested on the same scene, which can learn feature representations for a specific scene and HAD task. The learned feature representations tend to work poorly when directly transferred to other scenes or HAD tasks, exhibiting poor generalization.

In order to improve the generalization of DL-based methods, some scholars have introduced novel methods for HAD, which are trained in some scenes and transferred to other scenes. For example, Jiang et al. [41] proposed a weakly supervised discriminative learning method with spectrally constrained GAN for HAD (WeaklyAD). This method trains the model by the labeled samples obtained via coarse anomaly detection on a specific scene and transfers the trained model to the other scene obtained from the same sensor. Liu et al. [42] proposed a dual-frequency AE (DFAE) method, which introduces spectral domain learning to enhance the separability of backgrounds from anomalies. This method uses an HSI for training and transfers the trained model to other HSIs obtained from the same sensor for testing. Li et al. [43] proposed a general anomaly enhancement network with generative SSL (GSSL) for HAD, named AETNet, which was trained on a set of anomaly-free HSIs with block-level random masks. The best pretrained model was searched during the transfer learning stage by using global Mahalanobis distance to adapt the pretrained model to HAD tasks. In addition, AETNet is trained and tested on images of the same size, which requires complex window resizing when transferred to scenes of various sizes. According to the above analysis, while some progress has been made in boosting the generalization of DL-based HAD methods, there is still a pressing need for a DL-based HAD method that can not only perform pretraining without labeled sample generation but also flexibly transfer the pretrained model to various HAD datasets without the limitation of image sensor and image size.

Inspired by the SSL [44], we propose a contrastive SSL (CSSL)-based background reconstruction method for HAD, which can be labeled as contrastive self-supervised learning based background reconstruction method for HAD (CSSBR). The overall architecture of CSSBR consists of two parts: self-supervised pretraining and transferring the pretrained model to HAD tasks. First, the pretraining model is constructed via a pixel- and patch-level random masking strategy, a dual attention network (DAN) encoder, and a lightweight decoder. The pixel- and patch-level random masking strategy is designed to randomly mask off some pixels or patches of the original HSI to obtain pixel- and patch-level random masked image pairs with the same semantic information, which can help to complete the pretraining task. The DAN encoder based on vision transformer (ViT) and channel attention module (CAM) is constructed to extract the global context information and the correlation between spectra. A lightweight decoder is introduced to reconstruct the original image by reverse-mapping the encoded features and the masked patches. The pretraining model is trained by reconstructing the backgrounds from the patch-level random masked image and contrasting between pixel- and patch-level random masked images. Also, the loss function based on a designed background reconstruction guided (BRG) loss and a cross-entropy (CE) loss is introduced to guide pretraining to obtain background reconstruction without labeled sample generation and to learn general background representations for different downstream HAD tasks. Subsequently, the DAN encoder and the lightweight decoder are transferred to downstream tasks, and a simple fine-tuning of the lightweight decoder through BRG loss can obtain adaptive background reconstruction for various HAD tasks. Finally, anomaly detection is performed using reconstruction errors.

The main contributions of our research are summarized as follows.

1) A novel DAN encoder is proposed based on ViT and CAM, which can effectively extract the global context information and the correlation between spectra.
2) A pretraining model is constructed by a CSSL-based background reconstruction method, where a pixel- and patch-level random masking strategy is designed to construct background reconstruction task and contrastive learning task to learn general background representations of HSI without labeled sample generation.
3) A transfer learning model is constructed based on the DAN encoder and a lightweight decoder, where a BRG loss is proposed to fine-tune the transfer learning model to obtain adaptive background reconstruction without labeled sample generation for various HAD tasks.

The rest of this article is organized as follows. Section II briefly analyzes the related work. Section III describes the details of the proposed CSSBR method in detail. Section IV analyzes and discusses the experimental results. Finally, the conclusions are drawn in Section V.

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II. Related Work

A. Self-Supervised Learning

SSL learns general feature representations of large amounts of unlabeled data by designing a pretext task [45]. SSL has been a huge success not only in computer vision (CV) [46], [47], [48], [49], [50] but also in natural language processing (NLP) [51], [52], [53] and other areas. Most of the mainstream SSL methods can be divided into two types: generative and contrastive. GSSL learns representations by reconstructing or generating input data [54]. Masked image modeling (MIM) [55] is a typical pretraining method in GSSL that learns general and robust representations from unlabeled image data by masking parts of the input image and predicting the masked content. On the other hand, CSSL aims to learn the invariance of differently augmented data to obtain more discriminative feature representations [56], [57]. In recent years, some CSSL methods have been proposed in the CV field, such as momentum contrast (MoCo) for unsupervised visual representation learning [47], bootstrap your own latent (BYOL) [48], and knowledge distillation with no labels (DINO) [49]. MoCo uses a momentum encoder to generate positive and negative sample features and encode negative sample features as keys to be stored in a dynamic dictionary to facilitate contrastive learning. BYOL builds on MoCo by removing negative samples and achieving feature optimization through the interaction of two neural networks. DINO introduces knowledge distillation, uses the momentum average to update the teacher network through the student network, and uses the CE loss for training.

GSSL has been introduced into HAD fields such as PDBSNet [38], MSNet [39], SSHADFormer [40], and AETNet [43]. Among them, only AETNet constructs a pretraining model via GSSL based on MIM and transfers the pretrained model to different HAD tasks. CSSL has been successfully applied in the field of HSI processing, such as change detection [58], target detection [56], and image classification [59]. To the best of our knowledge, CSSL is less explored for HAD tasks. Inspired by current SSL methods, we integrate the MIM-based GSSL approach with the CSSL approach to design a CSSL-based background reconstruction method to construct the pretraining model. Unlike previous methods, our method implements image masking from pixel-level and patch-level to construct background reconstruction task and contrastive learning task. The contrastive learning task is utilized to help the background reconstruction task to obtain general background representations without labeled sample generation, which can be flexibly transferred to various HAD datasets without the limitation of image sensor and image size.

B. Transformer and ViT

Transformer is a DL model designed to process sequence data, proposed by Vaswani et al. [60]. Transformer was initially designed for NLP tasks, particularly achieving significant success in machine translation. Unlike traditional recurrent neural networks (RNNs) and CNNs, transformer is based on the self-attention (SA) mechanism that computes the relationship or dependency between each element in the sequence and all other elements in the sequence [61]. It can process all elements in the sequence in parallel, thereby improving training efficiency. The transformer encoder is composed of multiple identical layers stacked, each including a multihead SA (MSA) mechanism sublayer and a fully connected (FC) feedforward network sublayer. Residual connections are used around the two sublayers followed by layer normalization [60]. ViT [61] is the first model to successfully apply the transformer architecture to CV tasks. Unlike 1-D sequential data in NLP, the data in CV are usually 2-D or 3-D. Therefore, ViT first splits the image into nonoverlapping patches of fixed size. Then, these patches are flattened into 1-D vectors for linear embedding. At the same time, position embedding is added to preserve the relative positional relationships between the image patches, thus capturing the spatial relationships of the image. Finally, the generated embeddings are input into the transformer encoder.

ViT-based methods have been successfully applied in HSI processing, such as image classification [62], anomaly detection [43], and target detection [63]. Although ViT-based methods have achieved good performance in anomaly detection, they still face the problem of insufficient utilization of spectral information. We designed a novel DAN encoder based on ViT and CAM to improve the extraction of spectral features while learning spatial information. Besides, we construct a pretraining model based on DAN encoder to obtain general background representations of HSI.

III. Method

Our method can be divided into two parts: self-supervised pretraining and transferring the pretrained model to HAD tasks. The overall architecture of CSSBR is shown in Fig. 1.

A. Self-Supervised Pretraining

The pretraining model aims to learn general background feature representations of images by CSSL-based background reconstruction. The pretraining framework is divided into six parts, namely, pixel- and patch-level random masking strategy, teacher encoder, student encoder, student decoder, projection header, and loss function.

1) Pixel- and Patch-Level Random Masking Strategy: Data augmentation plays a crucial role in SSL and is the key to improving the performance of model. We design a pixel- and patch-level random masking strategy, including pixel-level random masking and patch-level random masking, to randomly mask off some pixels or patches of the original HSI. Thus, we can obtain image pairs with the same semantic information, which helps to complete the pretraining task. We denote the original pretraining HSI as $\boldsymbol{X} \in \mathbb{R}^{b\times h\times w}$, where b, h, and w represent the number of bands, height, and width, respectively. The pixel-level random masking employs a masking ratio of n to randomly mask some pixels of the original pretraining HSI, generating the pixel-level random masked image $\boldsymbol{X}t$. The number of total pixels is $N = h \times w$ and the number of masked pixels is $N' = n \times N$. Patch-level random masking first follows ViT to divide the

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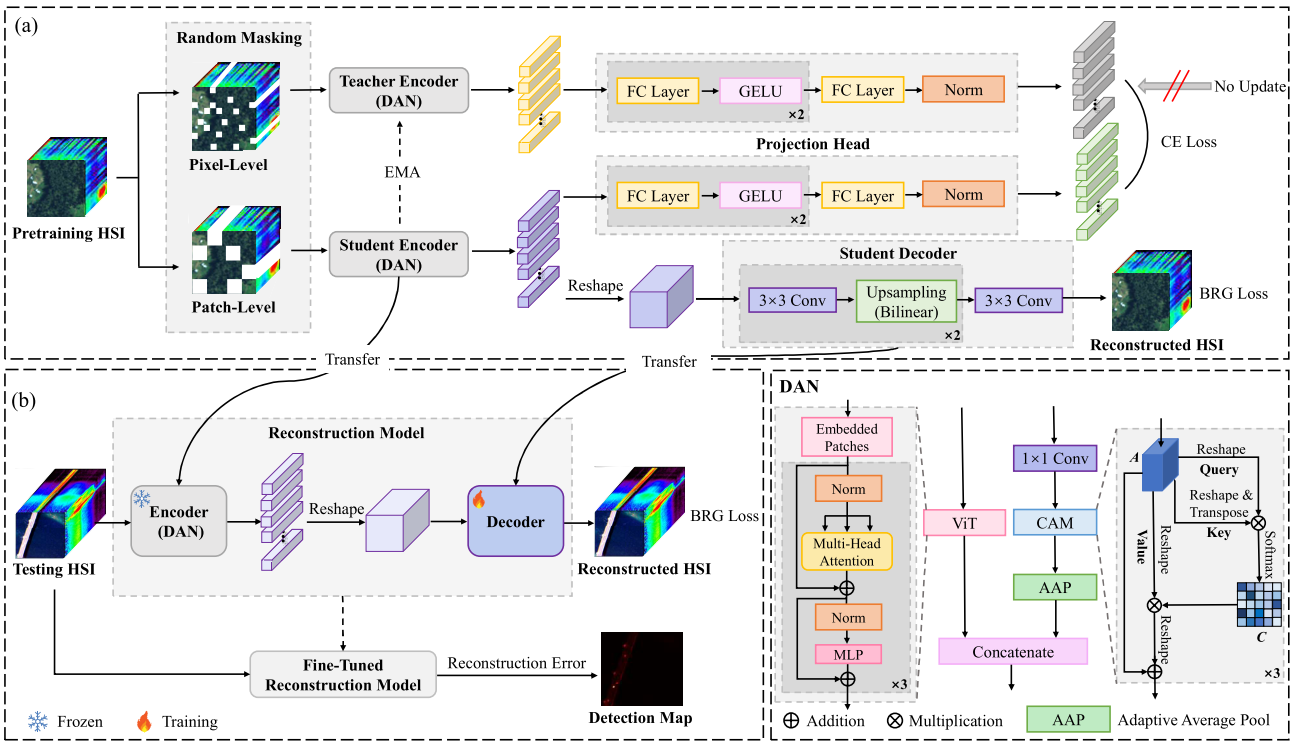


Fig. 1. Overall architecture of the proposed CSSBR method. (a) Pretraining. (b) Transferring to HAD.

original pretraining HSI into regular nonoverlapping patches and then employs a masking ratio of m to randomly mask some patches, generating the patch-level random masked image $\boldsymbol{X}s$. The size of patch is the same as the size of the patch divided by the ViT in the encoder, which is experimentally set as $p = 4$ in this work. The number of patches is $M = h/p \times w/p$ and the number of masked patches is $M' = m \times M$. Patch-level random mask images are used to construct the background reconstruction task, while pixel- and patch-level random mask images are used to construct the contrastive learning task. The contrastive learning task can help the background reconstruction task to obtain general background representations without labeled sample generation.

2) Encoder: It consists of two components, teacher encoder and student encoder. Two encoders maintain the same network architecture via DAN backbone on different masked images to learn feature representations. The network architecture of DAN is shown in Fig. 1. The network includes a spatial branch and a spectral branch to extract spatial and spectral features through ViT and CAM. The spatial branch focuses on extracting the global context information, and the spectral branch focuses on capturing the correlation between spectra.

The spatial branch uses a three-layer transformer with three heads. First, the input HSI is divided into nonoverlapping patches. Second, these patches are flattened and the flattened patches are mapped using linear projection to obtain patch embeddings. Finally, the full patch embedding is used as input to learn the latent feature representation and obtain a spatial feature vector.

The spectral branch consists of three CAMs, which first downscales the spectrum with band number of b by using a 1×1 convolutional layer to remove redundant information, generating a new spectrum with band number of b'. The correlation between the remaining spectral bands is then captured by three CAMs. In order to concatenate with the output spatial features of the spatial branch, an adaptive average pooling (AAP) layer operation is performed at the end of the spectral branch to obtain a spectral feature vector.

The structure of CAM is shown in Fig. 1. First, we reshape the input feature $\boldsymbol{A} \in \mathbb{R}^{b' \times h \times w}$ to $\mathbb{R}^{b' \times N}$, multiply matrix $\boldsymbol{A}$ with the transpose of $\boldsymbol{A}$, and input their result to a softmax layer. Thus, we obtain the channel attention map $\boldsymbol{C} \in \mathbb{R}^{b' \times b'}$, which can be expressed as follows:

$$c_{ji} = \frac{\exp\left(\boldsymbol{A}_i \boldsymbol{A}_j^{\mathrm{T}}\right)}{\sum_{i=1}^{b'} \exp\left(\boldsymbol{A}_i \boldsymbol{A}_j^{\mathrm{T}}\right)} \tag{1}$$

where c_{ji} measures the ith channel's impact on the jth channel and $\boldsymbol{A}_i$ and $\boldsymbol{A}_j$ are the ith and jth row of matrix $\boldsymbol{A}$, respectively. Then, we perform a matrix multiplication between the channel attention map $\boldsymbol{C}$ and the reshaped input feature $\boldsymbol{A}$ and reshape their result to $\mathbb{R}^{b' \times h \times w}$. Subsequently, we multiply the result by the learnable scale parameter β to perform an element-wise sum operation with the input feature $\boldsymbol{A}$ to obtain the final output

$$\boldsymbol{I}_j = \beta \sum_{i=1}^{b'} (c_{ji} \boldsymbol{A}_i) + \boldsymbol{A}_j \tag{2}$$

where $\boldsymbol{I}_j$ is the jth row of matrix $\boldsymbol{I}$ and represents the final feature of each channel, which can be expressed as a weighted sum of the features of all channels and the original features. In other words, CAM helps to capture the long-distance

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correlation between spectral features and improve spectral feature extraction.

Finally, the features obtained from the spatial and spectral branches are concatenated and the final feature is 320 dimensions, where the spatial and spectral branches output features with dimensions of 192 and 128, respectively.

3) Teacher Encoder: The teacher encoder takes pixel-level random masked images as input. The teacher encoder is constructed based on the state of the student encoder in the past iterations. A parameter update is performed between the teacher encoder and the student encoder via the exponential moving average (EMA), with update rule $\theta_t \leftarrow \lambda\theta_t + (1-\lambda)\theta_s$. θ_t and θ_s are the parameters of the teacher encoder and the student encoder, respectively. λ follows a cosine schedule from 0.9995 to 1 during the training period.

4) Student Encoder: The student encoder takes patch-level random masked images as input and is trained by cropping the input and inserting learnable embeddings. Thus, the student encoder can learn a potential feature representation of the masked image, which is then used to reconstruct the original signal of the masked patches. The trained student encoder will be transferred to HAD tasks as a feature extractor.

5) Student Decoder: The student decoder consists of three convolutional layers, where each layer contains a 2-D convolutional operation with a kernel size of 3×3 and a step size of 1. Bilinear interpolation is used for upsampling after the first two 2-D convolution operations to recover a reconstructed image of the same spatial resolution as the encoded image. The student decoder finally converts the encoded features back into the original feature space to reconstruct the original HSI from the masked image. Since the student decoder is a sequence of convolutional layers rather than ViT and CAM, the decoder is a lightweight structure in order to reduce computational consumption in pretraining and transferring learning.

6) Projection Head: The projection head consists of a three-layer multilayer perceptron (MLP), where the first two layers use the Gaussian error linear unit (GELU) activation function in order to introduce nonlinear transformations, and the last layer is a linear layer without activation function but with an L_2 normalization layer and a weight normalized FC layer with Z dimensions. The projection head is added after both the student encoder and the teacher encoder to map the features output from two encoders into a space suitable for contrastive learning. The projection head behind the teacher's encoder is also updated by EMA.

We denote student encoder, student decoder, and teacher encoder as $E\theta_s$, $D\theta_s$, and $E\theta_t$, respectively. For the input of pixel-level random masked image $\boldsymbol{X}t$ and patch-level random masked image $\boldsymbol{X}s$, both encoder networks finally output probability distributions on Z dimensions named P_s and P_t, which can be expressed as follows:

$$P_s(\boldsymbol{X}s)^{(i)} = \frac{\exp(D\theta_s(E\theta_s(\boldsymbol{X}s)^{(i)})/\tau_s)}{\sum_{z=1}^{Z}\exp(D\theta_s(E\theta_s(\boldsymbol{X}s)^{(z)})/\tau_s)} \tag{3}$$

$$P_t(\boldsymbol{X}t)^{(i)} = \frac{\exp(E\theta_t(\boldsymbol{X}t)^{(i)}/\tau_t)}{\sum_{z=1}^{Z}\exp(E\theta_t(\boldsymbol{X}t)^{(z)}/\tau_t)} \tag{4}$$

where the temperature parameter τ is commonly used in knowledge distillation to control the clarity of the output distribution. The dimension Z of the probability distributions output by the encoder networks is set to 1024.

7) Loss Function: Considering that the pretraining framework involves contrastive learning task and reconstruction task, the widely used CE loss and mean square error (mse) loss can be integrated to train the pretraining model. The CE loss can constrain the training of contrastive learning network by aligning the final output probability distributions of the teacher encoder and the student encoder to learn discriminative features. The mse loss can constrain the training of reconstruction network to ensure image reconstruction quality. With the integration of CE loss and mse loss, the pretraining framework can capture not only discriminative features between masked image pairs with the same semantic information but also overall features of the patch-level masked image, thus achieving good feature representations of HSI. However, the learned feature representations of the whole image are not enough for HAD tasks, where we usually extract background features to distinguish anomalies from backgrounds. Inspired by total variation (TV) loss [64], we propose a BRG loss based on mse loss and TV loss to boost the background feature learning by guiding the training of reconstruction network to strengthen background reconstruction.

The TV loss considers the difference between the pixels and their surrounding pixels in the image, aiming to smooth the image by reducing local variations between pixels. The background is dominant in terms of the pixels number and there is less spectral feature variation between neighboring pixels within the same background category. In contrast, the anomalies are usually smaller and sparser, showing marked spectral feature differences from the surrounding background. Based on the above facts, the BRG loss can enhance the smoothness of the reconstructed background by constraining the differences between neighboring pixels. Thereby, the BRG loss can improve the reconstruction network's ability to extract background features.

The BRG loss can be calculated as follows:

$$L_{\text{mse}} = \frac{1}{hw}\sum_{i=1}^{h}\sum_{j=1}^{w}(\boldsymbol{x}_{i,j} - \tilde{\boldsymbol{x}}_{i,j})^{\mathrm{T}}(\boldsymbol{x}_{i,j} - \tilde{\boldsymbol{x}}_{i,j}) \tag{5}$$

$$L_{\text{TV}} = \frac{1}{h(w-1)b}\sum_{i=1}^{h}\sum_{j=1}^{w-1}\left|\tilde{\boldsymbol{x}}_{i,j} - \tilde{\boldsymbol{x}}_{i,j+1}\right| + \frac{1}{(h-1)wb}\sum_{i=1}^{h-1}\sum_{j=1}^{w}\left|\tilde{\boldsymbol{x}}_{i,j} - \tilde{\boldsymbol{x}}_{i+1,j}\right| \tag{6}$$

$$L_{\text{BRG}} = L_{\text{mse}} + L_{\text{TV}} \tag{7}$$

where $\boldsymbol{x}_{i,j} \in \mathbb{R}^{b\times 1}$ and $\tilde{\boldsymbol{x}}_{i,j} \in \mathbb{R}^{b\times 1}$ denote the spectral vectors of pixel with spatial position (i, j) in the original pretraining HSI and the reconstructed pretraining HSI, respectively. L_{mse} represents the mse loss between the original HSI and the reconstructed HSI. L_{TV} represents the TV loss, which calculates the average difference between the pixels and their surrounding pixels in the reconstructed image.

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The CE loss and BRG loss constitute the total loss for the pretraining phase, which is calculated as follows:

$$L_{\mathrm{CE}} = \min_{\theta_s} H(P_t(\boldsymbol{X}t), P_s(\boldsymbol{X}s)) \tag{8}$$

$$L_{\mathrm{Total}} = L_{\mathrm{BRG}} + L_{\mathrm{CE}} \tag{9}$$

where $H(a, b) = -a \log b$ and L_{CE} represents the CE loss, which calculates the difference between the output probability distributions of student encoder and teacher encoder.

B. Transfer Learning

This section describes how to efficiently transfer the pretrained model to various HAD tasks without labeled samples. The transfer learning model contains four parts, namely, encoder, fine-tuning decoder, loss function, and anomaly detection.

1) Encoder: We transfer the pretrained student encoder to downstream HAD task. Feature extraction is performed on the testing image via the pretrained student encoder, producing low-dimensional, abstract, and general background representation. We denote the testing HSI as $\boldsymbol{Y} \in \mathbb{R}^{B\times H\times W}$, where B, H, and W represent the number of bands, height, and width, respectively.

2) Fine-Tuning Decoder: To reconstruct the background for the testing image, we transfer the student decoder to downstream HAD task to reverse-map the encoded features into the original feature space. During the fine-tuning process, the encoder parameters are not updated, which can obtain a general background representation and improve the efficiency of fine-tuning. Due to the limited pretraining images, the data distribution of the downstream image may be different from that of the pretraining images. Therefore, the decoder should be continuously updated based on the parameters obtained from pretraining according to the testing image, which makes the model better adapted to the specific HAD task and improves the ability of background reconstruction.

3) Loss Function: In order to enable the reconstruction model to better reconstruct the background of a specific testing image, we perform a simple fine-tuning through the BRG loss to efficiently transfer the pretrained encoder and the lightweight decoder to a variety of HAD tasks without labeled samples.

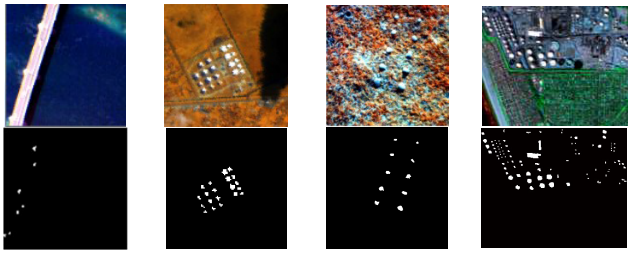
4) Anomaly Detection: Once the reconstruction model is fine-tuned through BRG loss, each testing image is fed into the pretrained encoder and the fine-tuned decoder, producing a reconstructed background image. If the tested pixel is background, the reconstruction error between the testing image and the reconstructed image is likely to be small; otherwise, the reconstruction error is likely to be large. The anomaly detection result can be obtained as follows:

$$D_{i,j} = \left\| \boldsymbol{y}_{i,j} - \tilde{\boldsymbol{y}}_{i,j} \right\|_2 \tag{10}$$

where $\boldsymbol{y} \in \mathbb{R}^{B\times 1}$ and $\tilde{\boldsymbol{y}} \in \mathbb{R}^{B\times 1}$ denote the spectral vectors of the tested pixel with spatial position (i, j) in the testing image and its reconstructed image, respectively; and $D_{i,j}$ denotes the anomaly probability of the tested pixel, which ultimately constitutes the anomaly detection map.



Fig. 2. Partial images of pretraining dataset HAD100.



(a) (b) (c) (d)

Fig. 3. Pseudo-color images and ground-truth maps of the testing datasets. (a) Pavia. (b) Texas Coast. (c) Cri. (d) EI Segundo.

IV. Experiment

A. Experiment Details

1) Experimental Datasets: In this work, one hyperspectral dataset HAD100 with a large scale is used for pretraining, and four hyperspectral datasets, Pavia, Texas Coast, Cri, and EI Segundo, are used for downstream HAD tasks. Details of each dataset are given as follows.

1) *HAD100 Dataset [43]:* It was acquired by the Airborne Visible/Infrared Imaging Spectrometer-Next Generation (AVIRIS-NG) sensor with a total of 100 images, some of which are shown in Fig. 2. The images were uniformly cropped into patches of size 64 × 64 with a spectral range of 377–2500 nm. Original band 425 and 276 bands reserved. Considering that the dataset includes not only type-rich background features such as grasslands, deserts, and lakes but also different anomalous targets such as vehicles, boats, and buildings, the dataset is used for pretraining.
2) *Pavia Dataset [17]:* It was acquired over the northern Italy city of Pavia using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. Fig. 3(a) shows the pseudo-color image and the ground-truth map. The data size is 108 × 120 pixels with a spatial resolution of 1.3 m. The spectral coverage spans from 430 to 860 nm, encompassing 102 bands after the elimination of noise bands and water vapor absorption bands. The anomalies are vehicles and piers on the bridge, consisting of 43 anomalous pixels.
3) *Texas Coast Dataset [65]:* It was acquired over the coastal region of Texas in the United States using the AVIRIS sensor. Fig. 3(b) shows the pseudo-color image and the ground-truth map. The data size is 100 × 100 pixels with a spatial resolution of 17.2 m. The spectral coverage spans from 450 to 1350 nm,

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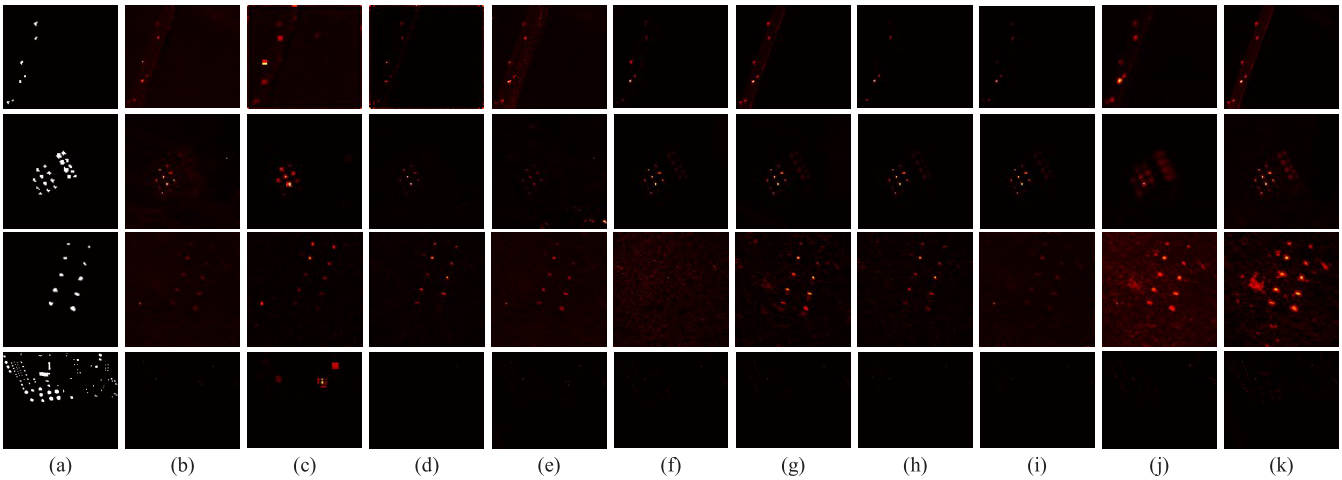


Fig. 4. Detection maps of different methods. (a) Ground-truth maps of Pavia, Texas Coast, Cri, and EI Segundo datasets. (b) GRX. (c) 2S-GLRT. (d) CRD. (e) FEBPAD. (f) Auto-AD. (g) GAED. (h) PDBSNet. (i) MSNet. (j) AETNet. (k) CSSBR.

encompassing 207 bands. The anomalies are buildings, consisting of 155 anomalous pixels.

4) *Cri Dataset [66]:* It was acquired by the Nuance Cri hyperspectral sensor. Fig. 3(c) shows the pseudo-color image and the ground-truth map. The size of the data is 400 × 400 pixels. The spectral coverage spans from 650 to 1100 nm, encompassing 46 bands. The anomalies are stones, consisting of 1254 anomalous pixels.
5) *EI Segundo Dataset [67]:* It was acquired by the AVIRIS sensor in the EI Segundo area, CA, USA. Fig. 3(d) shows the pseudo-color image and the ground-truth map. The data size is 250 × 300 pixels with a spatial resolution of 7.1 m. The spectral coverage spans from 366 to 2496 nm, encompassing 224 bands. The anomalies are tanks and towers in the oil refinery, consisting of 2048 anomalous pixels.

2) Experimental Settings: To validate the effectiveness of the proposed method CSSBR, we compare it with nine HAD methods, including four traditional methods, GRX [8], 2S-GLRT [13], CRD [14], and FEBPAD [23], and five DL methods, Auto-AD [34], GAED [35], PDBSNet [38], MSNet [39], and AETNet [43]. Through experiments and referencing the original paper, all the comparison methods used the optimal parameters.

For the pretraining network in the CSSBR method, we use the AdamW optimizer and train for 100 epochs, with the learning rate increasing linearly to 0.0001 within the first ten epochs, followed by decaying the learning rate using a cosine schedule. Temperature τ_s is 0.1, and a cosine warm-up from 0.04 to 0.07 is used for τ_t within the first 30 epochs. The batch size and patch size are set to 16 and 4, respectively. The patch-level random masking ratio and the pixel-level random masking ratio are both set to 90% for all datasets.

In the fine-tuning network, the AdamW optimizer is still utilized with a learning rate of 0.0001 and 200 epochs. The patch size is still set to 4. All experiments are based on the PyTorch framework in the 64-bit Windows 10 operating system, NVIDIA GeForce RTX3060 graphics processing unit (GPU) with 16 GB of memory.

3) Evaluation Metrics: In order to assess the anomaly detection performance of different methods, we use a variety of evaluation metrics, namely, detection map, receiver operating characteristic (ROC) curve, area under the curve (AUC), separability map, and run time.

The detection map is segmented using different thresholds (τ) to obtain different false alarm rates (P_f) and probability of detection (P_d). We use the ROC curves of (P_f, P_d) and (τ, P_d) to evaluate the effectiveness of anomaly detector and the anomaly detection ability, respectively. The closer the ROC curve of (P_f, P_d) is to the top-left corner, the better the effectiveness of anomaly detector. The closer the ROC curve of (τ, P_d) is to the top-right corner, the higher the anomaly detection ability.

The AUC value indicates the area under the ROC curve, and we use a variety of AUC values for overall evaluation [68], [69]. Specifically, higher values of $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ indicate better detector effectiveness. Higher values of $\mathrm{AUC}_{(\tau,P_\mathrm{d})}$ indicate better anomaly detection ability. Lower values of $\mathrm{AUC}_{(\tau,P_\mathrm{f})}$ indicate better background suppression performance. $\mathrm{AUC}_{\mathrm{JAD}}$ is $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}+\mathrm{AUC}_{(\tau,P_\mathrm{d})}$, which further indicates the anomaly detection ability. $\mathrm{AUC}_{\mathrm{ADBS}}$ is $\mathrm{AUC}_{(\tau,P_\mathrm{d})}-\mathrm{AUC}_{(\tau,P_\mathrm{f})}$, which indicates the anomaly detection and background suppression performance. $\mathrm{AUC}_{\mathrm{OAD}}$ is the sum of $\mathrm{AUC}_{(\tau,P_\mathrm{d})}$ and $1-\mathrm{AUC}_{(\tau,P_\mathrm{f})}$, which is the second $\mathrm{AUC}_{\mathrm{ADBS}}$ metric derived from the classification point of view and indicates the overall anomaly detection performance. $\mathrm{AUC}_{\mathrm{SBPR}}$ is computed by $\mathrm{AUC}_{(\tau,P_\mathrm{d})}/(1-\mathrm{AUC}_{(\tau,P_\mathrm{f})})$, which is also an effective detection metric derived from a widely used idea in communications or signal processing.

B. Detection Performance Analysis

In this section, we compare the detection performance of the proposed CSSBR method with other methods on four different hyperspectral datasets. To mitigate the effects of random factors, all experiments were run five times consecutively. The value of the third-ranked $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ and its corresponding

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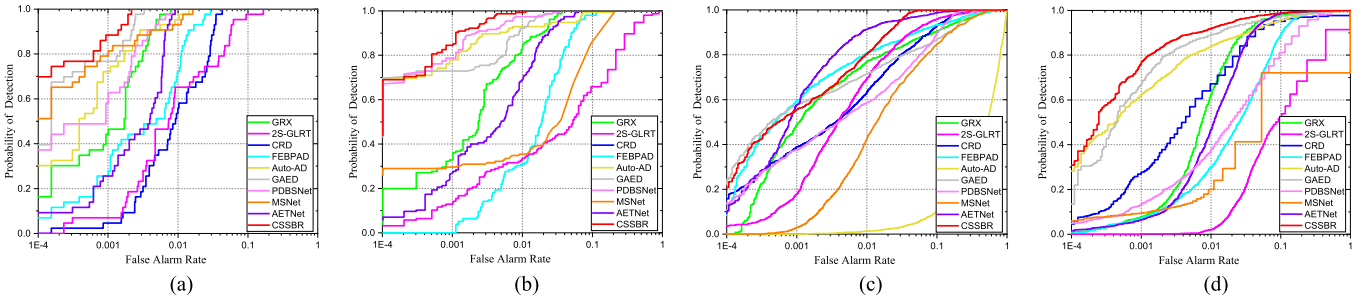


Fig. 5. ROC curves of (P_f, P_d) for different methods on four datasets. (a) Pavia. (b) Texas Coast. (c) Cri. (d) EI Segundo.

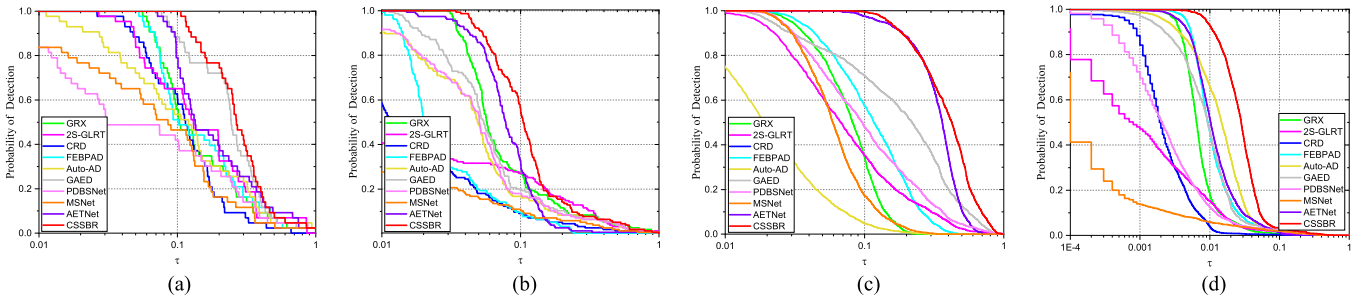


Fig. 6. ROC curves of (τ, P_d) for different methods on four datasets. (a) Pavia. (b) Texas Coast. (c) Cri. (d) EI Segundo.

values of other evaluation metrics were taken as the final results. The experiment's purpose is to demonstrate the effectiveness of our pretrained model in transferring to various HAD tasks.

Fig. 4 shows the detection maps of different methods on four datasets. The results show that CSSBR is able to detect all the anomalies and highlight the anomalies in the first three datasets, for example, the right anomalies in Texas Coast dataset and the top-right and bottom-right anomalies in the Cri dataset are well detected and highlighted. Besides, CSSBR is able to detect more anomalies in the EI Segundo dataset in comparison to the other methods.

Fig. 5 shows the ROC curves of (P_f, P_d) for different methods on four datasets. On Pavia, Texas Coast, and EI Segundo datasets, the ROC curves of (P_f, P_d) for CSSBR are essentially located above that of the other methods. On the Cri dataset, CSSBR has the highest probability of detection when the false alarm rate is greater than 0.03. The results generally show that CSSBR exhibits a better effectiveness of detector.

Fig. 6 shows the ROC curves of (τ, P_d) for different methods on four datasets. On all four datasets, the ROC curves of (τ, P_d) for CSSBR are basically located above that of the other methods. The results show that CSSBR exhibits a higher anomaly detection ability than other comparison methods on four datasets.

Table I shows the AUC values for different methods on four datasets, where the optimal values are shown in bold. CSSBR was optimal for most AUC values in all datasets. Specifically, the $AUC_{(P_f,P_d)}$ values of CSSBR are the highest on all datasets, validating the effectiveness of the proposed method in anomaly detection. The $AUC_{(\tau,P_d)}$ and AUC_{JAD} values of CSSBR are the highest on all the datasets, indicating that the proposed method has a strong anomaly detection ability. Even though $AUC_{(\tau,P_f)}$ values of CSSBR are not optimal, the AUC_{ADBS}, AUC_{OAD}, and AUC_{SBPR} values of CSSBR are the highest in all the datasets, indicating that the proposed method is able to achieve better overall results in highlighting the anomalies and suppressing the background. In summary, in terms of various AUC values, CSSBR is generally superior to other methods.

Fig. 7 shows the separability maps for different methods on four datasets, which can visually reflect the separation of anomalies and backgrounds. On all the datasets, CSSBR exhibits larger spacing between the anomaly box and the background box compared to the other methods, indicating that CSSBR is able to separate anomalies from backgrounds well. In addition, the center line of the anomaly box of CSSBR has a higher position and the bottom of the anomaly box is higher on all datasets compared to the other methods, indicating that CSSBR highlights the anomalies well. In summary, in terms of separability maps, CSSBR outperforms other methods in distinguishing anomalies from backgrounds.

Table II shows the run times of different methods on four datasets, where the run time represents the total time of performing the downstream task. The average run time indicates the time efficiency of each method on various datasets. Compared to traditional methods 2S-GLRT and CRD, CSSBR requires a shorter average runtime. Compared to all DL-based methods, CSSBR requires the shortest average run time. Due to the freezing of the encoder, the number of parameters in the downstream task is significantly reduced, thus improving the time efficiency of the CSSBR method.

By comparing the proposed CSSBR method and the other methods on four different hyperspectral datasets, in terms of detection map, ROC curve, AUC value, separability

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TABLE I
AUC VALUES FOR DIFFERENT METHODS ON FOUR DATASETS

		GRX	2S-GLRT	CRD	FEBPAD	Auto-AD	GAED	PDBSNet	MSNet	AETNet	CSSBR
Pavia	$AUC_{(P_f,P_d)}$	0.9982	0.9787	0.9859	0.9933	0.9984	0.9994	0.9985	0.9985	0.9963	**0.9996**
	$AUC_{(\tau,P_f)}$	0.0231	0.0249	0.0055	0.0174	0.0010	0.0044	0.0004	**0.0002**	0.0093	0.0072
	$AUC_{(\tau,P_d)}$	0.1880	0.2106	0.1416	0.1962	0.1864	0.2799	0.1491	0.1333	0.2423	**0.3080**
	AUC_{JAD}	1.1861	1.1893	1.1275	1.1895	1.1848	1.2793	1.1477	1.1318	1.2386	**1.3076**
	AUC_{ADBS}	0.1649	0.1857	0.1362	0.1787	0.1853	0.2755	0.1488	0.1332	0.2330	**0.3008**
	AUC_{OAD}	1.1649	1.1857	1.1362	1.1787	1.1853	1.2755	1.1488	1.1332	1.2330	**1.3008**
	AUC_{SBPR}	0.1924	0.2160	0.1424	0.1996	0.1866	0.2811	0.1492	0.1334	0.2445	**0.3103**
Texas Coast	$AUC_{(P_f,P_d)}$	0.9945	0.8903	0.9574	0.9763	0.9947	0.9975	0.9986	0.9529	0.9908	**0.9994**
	$AUC_{(\tau,P_f)}$	0.0135	0.0020	0.0051	0.0072	0.0013	0.0031	0.0007	**0.0001**	0.0051	0.0093
	$AUC_{(\tau,P_d)}$	0.1147	0.0887	0.0481	0.0414	0.0904	0.0991	0.0925	0.0421	0.0932	**0.1454**
	AUC_{JAD}	1.1093	0.9790	1.0055	1.0176	1.0851	1.0966	1.0911	0.9950	1.0841	**1.1448**
	AUC_{ADBS}	0.1012	0.0868	0.0431	0.0342	0.0891	0.0960	0.0918	0.0420	0.0881	**0.1361**
	AUC_{OAD}	1.1012	1.0868	1.0431	1.0342	1.0891	1.0960	1.0918	1.0420	1.0881	**1.1361**
	AUC_{SBPR}	0.1163	0.0889	0.0484	0.0417	0.0906	0.0994	0.0926	0.0421	0.0937	**0.1467**
Cri	$AUC_{(P_f,P_d)}$	0.9675	0.9805	0.9763	0.9752	0.4931	0.9659	0.9691	0.9469	0.9939	**0.9942**
	$AUC_{(\tau,P_f)}$	0.0236	**0.0101**	0.0113	0.0249	0.0319	0.0156	0.0131	0.0219	0.0569	0.0326
	$AUC_{(\tau,P_d)}$	0.0869	0.1180	0.1460	0.1381	0.0316	0.2556	0.1546	0.0715	0.3558	**0.4284**
	AUC_{JAD}	1.0544	1.0985	1.1222	1.1134	0.5247	1.2215	1.1237	1.0184	1.3497	**1.4226**
	AUC_{ADBS}	0.0632	0.1079	0.1347	0.1132	-0.0003	0.2400	0.1415	0.0495	0.2989	**0.3957**
	AUC_{OAD}	1.0632	1.1079	1.1347	1.1132	0.9997	1.2400	1.1415	1.0495	1.2989	**1.3957**
	AUC_{SBPR}	0.0890	0.1192	0.1477	0.1417	0.0326	0.2596	0.1567	0.0731	0.3773	**0.4428**
EI Segundo	$AUC_{(P_f,P_d)}$	0.9840	0.8308	0.9642	0.9556	0.9822	0.9877	0.9450	0.8377	0.9812	**0.9943**
	$AUC_{(\tau,P_f)}$	0.0017	0.0030	0.0004	0.0032	0.0010	0.0005	0.0003	**0.0001**	0.0013	0.0028
	$AUC_{(\tau,P_d)}$	0.0118	0.0071	0.0043	0.0173	0.0234	0.0179	0.0098	0.0066	0.0168	**0.0366**
	AUC_{JAD}	0.9958	0.8378	0.9685	0.9730	1.0055	1.0056	0.9549	0.8443	0.9979	**1.0309**
	AUC_{ADBS}	0.0101	0.0040	0.0039	0.0142	0.0224	0.0174	0.0095	0.0065	0.0154	**0.0338**
	AUC_{OAD}	1.0101	1.0040	1.0039	1.0142	1.0224	1.0174	1.0095	1.0065	1.0154	**1.0338**
	AUC_{SBPR}	0.0118	0.0071	0.0043	0.0174	0.0234	0.0179	0.0098	0.0066	0.0168	**0.0367**

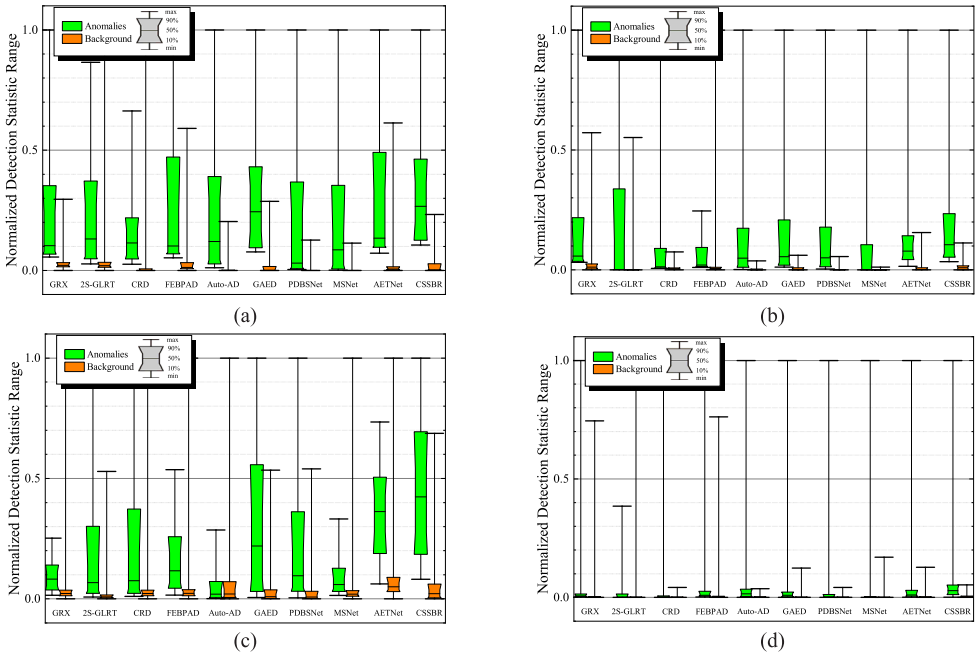


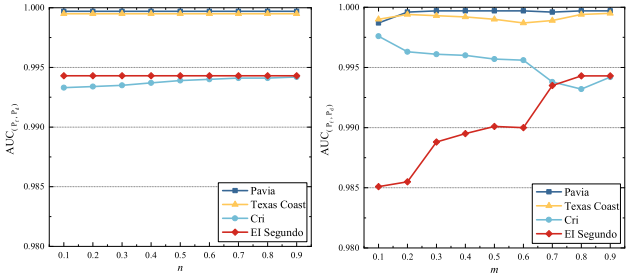
Fig. 7. Separability maps for different methods on four datasets. (a) Pavia. (b) Texas Coast. (c) Cri. (d) EI Segundo.

maps, and run time, we can draw a general conclusion that CSSBR outperforms the comparison methods, validating the effectiveness of transferring the pretrained model to various HAD tasks.

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TABLE II
RUN TIMES (IN SECONDS) FOR DIFFERENT METHODS ON FOUR DATASETS

Datasets	GRX	2S-GLRT	CRD	FEBPAD	Auto-AD	GAED	PDBSNet	MSNet	AETNet	CSSBR
Pavia	0.0430	16.7853	35.3500	1.7280	9.5300	28.3742	172.8967	304.7908	134.9112	2.1250
Texas Coast	0.0619	25.6274	4.7985	2.8221	4.0665	30.1980	175.5161	233.5868	140.3916	2.0390
Cri	0.2844	3167.1337	1057.5035	10.2299	91.4239	406.5747	942.5127	4420.2789	191.5660	33.6560
EI Segundo	0.4490	1813.9579	649.5329	23.0504	14.4946	304.7728	722.4207	2081.7926	173.4303	16.0396
Average	0.2096	1255.8761	436.7962	9.4576	29.8788	192.4799	503.3366	1760.1123	160.0748	13.4649



(a) (b)

Fig. 8. Parameter analysis on four datasets. (a) Pixel-level and (b) patch-level random masking ratio.

C. Parameter Analysis

In this section, we discuss the impact of two important parameters of the CSSBR method on the HAD performance. The two important parameters are pixel-level random masking ratio n and patch-level random masking ratio m. The experiments are conducted on four datasets with the evaluation metric of $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ value.

First, the pixel-level random masking ratio n was set according to the range [10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%], and the $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values for the four datasets are shown in Fig. 8(a). On Pavia, Texas Coast, and EI Segundo datasets, the $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values hardly change as n increases. On the Cri dataset, the $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values show a slowly increasing trend with increasing n, and the best detection result is achieved at $n = 90\%$. Considering the overall performance of the four datasets, we set the pixel-level random masking ratio n as 90%.

Subsequently, the patch-level random masking ratio m was set according to the range [10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%], and the $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values for the four datasets are shown in Fig. 8(b). On Pavia, Texas Coast, and EI Segundo datasets, the $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values show an overall increasing trend as m increases. On the Cri dataset, $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values show a decreasing and then increasing trend as m increases, but the change of $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values does not exceed 0.45%. Considering the overall performance of the four datasets, we set m to 90%.

D. Ablation Study

This section discusses the effectiveness of self-supervised pretraining, DAN, and BRG loss.

First, we discuss the effectiveness of self-supervised pretraining. Specifically, we compare the performance of three cases: 1) without using the pretraining model (Un-Pretrained); 2) using the pretraining model based on the MIM-based GSSL approach (GSSL-Pretrained); and 3) using the proposed self-supervised pretraining model (CSSBR-Pretrained). We aim to verify that the proposed pretraining model integrating the MIM-based GSSL approach and the CSSL approach enables CSSBR to learn general background representations for various downstream HAD tasks. The $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values and run times are shown in Table III, with the optimal value in bold. With the proposed pretraining model, the $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values are better on most datasets, especially on Cri dataset. The run times are reduced on all datasets compared to the case without using the pretraining model. The experimental results illustrate that the general background representations learned by the proposed pretraining model can be quickly and efficiently transferred to various HAD tasks.

TABLE III
ABLATION STUDIES ON PRETRAINING PHASE

Case	Metric	Pavia	Texas Coast	Cri	EI Segundo
Un-Pretrained	$AUC_{(P_f,P_d)}$	0.9996	0.9993	0.9515	0.9942
	Run time/s	3.6561	3.1245	111.3698	32.8088
GSSL-Pretrained	$AUC_{(P_f,P_d)}$	0.9996	0.9992	0.9892	**0.9944**
	Run time/s	3.6814	3.2231	**33.3273**	**14.3008**
CSSBR-Pretrained	$AUC_{(P_f,P_d)}$	**0.9996**	**0.9994**	**0.9942**	0.9943
	Run time/s	**2.1250**	**2.0390**	33.6560	16.0396

Subsequently, we discuss the effectiveness of the DAN. Specifically, we compare the performance of CAM, ViT, and DAN. The $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values are shown in Table IV. DAN has better $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values on most datasets compared to CAM or ViT. The results show that DAN based on CAM and ViT can indeed improve the feature extraction effect and enhance the anomaly detection performance.

Finally, we discuss the effectiveness of the BRG loss. Specifically, we use either the BRG loss or the mse loss to guide the training of the reconstruction networks in both the pretraining phase and the transferring learning phase. The $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values are shown in Table IV, with the optimal value in bold. The $\mathrm{AUC}_{(P_\mathrm{f},P_\mathrm{d})}$ values are improved by 0.01%, 0.12%, and 0.46% on the Texas Coast, Cri, and EI Segundo datasets, respectively. The results show that BRG loss improves the anomaly detection performance.

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TABLE IV
ABLATION STUDIES ON DIFFERENT COMPONENTS

Dataset	CAM	ViT	L_{BRG}	$AUC_{(P_f,P_d)}$
Pavia	√	-	-	0.9996
	-	√	-	0.9996
	√	√	-	0.9996
	√	√	√	**0.9996**
Texas Coast	√	-	-	0.9984
	-	√	-	0.9993
	√	√	-	0.9993
	√	√	√	**0.9994**
Cri	√	-	-	0.9875
	-	√	-	0.9936
	√	√	-	0.9930
	√	√	√	**0.9942**
El Segundo	√	-	-	0.9877
	-	√	-	0.9814
	√	√	-	0.9897
	√	√	√	**0.9943**

V. CONCLUSION

In this article, a CSSBR method is proposed for HAD. By constructing a self-supervised pretraining model based on a pixel- and patch-level masking strategy and a DAN encoder, the pretraining model can learn general background representation without labeled sample generation. The DAN encoder is constructed by ViT and CAM to extract the global context information and the correlation between spectra. The transfer learning model is constructed based on a DAN encoder and a lightweight decoder, which is simply fine-tuned through BRG loss and efficiently transferred to a variety of HAD tasks. Experiments on four challenging hyperspectral datasets validate that the CSSBR method is capable of performing various anomaly detection tasks with leading detection performance and high detection efficiency.

However, the proposed CSSBR method is still deficient in background suppression. This may be due to the limited number of pretraining images and the deviation of the background distribution between the downstream dataset and the pretraining dataset, resulting in limited background suppression performance. We will continue to investigate how to improve the background suppression performance.

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